TASK 1 — FULL CI + DOCKER BUILD + PUSH (GHCR)

🎯 **Scenario**

- Build Node.js app
- Create Docker image
- Push to GitHub Container Registry (GHCR)

```yaml
name: 01-CI-Docker-GHCR

on:
  push:
    branches: [main]

permissions:
  contents: read
  packages: write

jobs:
  build-and-push:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Setup Node
        uses: actions/setup-node@v4
        with:
          node-version: 18

      - name: Install Dependencies
        run: npm install

      - name: Run Tests
        run: npm test

      - name: Build App
        run: npm run build
```

```
    - name: Login to GHCR
      uses: docker/login-action@v3
      with:
        registry: ghcr.io
        username: ${{ github.actor }}
        password: ${{ secrets.GITHUB_TOKEN }}

    - name: Build & Push Docker Image
      uses: docker/build-push-action@v5
      with:
        push: true
        tags: ghcr.io/${{ github.repository }}:latest
```

🚀 🔥 TASK 2 — AWS DEPLOYMENT USING OIDC (SECURE)

🎯 Scenario

- No static AWS keys
- Use OIDC for secure login
- Deploy to S3

```
name: 02-OIDC-AWS-Deploy

on:
  push:
    branches: [main]

permissions:
  id-token: write
  contents: read

jobs:
  deploy:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout
        uses: actions/checkout@v4
```

```
    - name: Configure AWS Credentials (OIDC)
      uses: aws-actions/configure-aws-credentials@v4
      with:
        role-to-assume: arn:aws:iam::123456789012:role/github-role
        aws-region: us-east-1

    - name: Upload to S3
      run: aws s3 sync ./dist s3://my-bucket
```

🚀 🔥 TASK 3 — SONARQUBE + QUALITY GATE PIPELINE

🎯 Scenario

- Run code quality scan
- Fail pipeline if quality fails

```
name: 03-Code-Quality

on:
  pull_request:
    branches: [main]

jobs:
  quality:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v4

      - name: Setup Java (Sonar requirement)
        uses: actions/setup-java@v4
        with:
          distribution: temurin
          java-version: 17

      - name: Build Project
        run: mvn clean install

      - name: SonarQube Scan
```

```
      uses: sonarsource/sonarqube-scan-action@v2
      with:
        args: >
          -Dsonar.projectKey=my-project
          -Dsonar.host.url=${{ secrets.SONAR_HOST }}
          -Dsonar.login=${{ secrets.SONAR_TOKEN }}
```

🚀 🔥 TASK 4 — ENVIRONMENT + APPROVAL + DEPLOY

🎯 Scenario

- Deploy only after approval
- Use GitHub environment

```
name: 04-Prod-Deployment

on:
  push:
    branches: [main]

jobs:
  deploy:
    runs-on: ubuntu-latest
    environment: production

    steps:
      - uses: actions/checkout@v4

      - name: Install App
        run: npm install

      - name: Deploy App
        run: |
          echo "Deploying to production server"
          # Example:
          # scp -r ./dist user@server:/var/www/app
```

💡 Approval configured in GitHub UI

🚀 🔥 TASK 5 — SECURITY PIPELINE (TRIVY + DEP SCAN)

🎯 Scenario

- Scan Docker image for vulnerabilities
- Fail if critical issues

```yaml
name: 05-Security-Scan

on:
  push:
    branches: [main]

jobs:
  security:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v4

      - name: Build Docker Image
        run: docker build -t myapp:latest .

      - name: Scan with Trivy
        uses: aquasecurity/trivy-action@v0.20.0
        with:
          image-ref: myapp:latest
          severity: CRITICAL,HIGH
          exit-code: 1
```